SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering.. ASSESSMENT TOOL 1- APPLICATION BASED QUESTIONS

Assessment **Tool** 1 - Application BasedQuestions

Course Code: **CSA06**

Course Title**:** Design and Analysis of Algorithms

CO Assessed: CO1 - Precision (BL1, BL2, BL3**)**

Assessment:

Weightage:

Total Marks:

CO1 Statement:

Student Name:

Section / Batch:

40% of CIA

100 (2 Questions **x** **50** Marks**)**

Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)

Shri Harshini Devi. A **Reg. No.:**

Date:

192524104

09/0712026

Instructions to Students

Answer BOTH questions. Each question carries *50* Marks **Total:** ***100*** Marks. Clearly show all steps in derivations and complexity analysis. Use proper asymptotic notation (Big-O, Big-O, Big-**)** wherever required. Justify each step in recurrence solving using appropriate methods. Neat presentation and logical explanation are mandatory

Question Type

Application-Based Questions

Focus Area

Questions

Marks

Apply Master Theorem and asymptotic analysis to real-world scenarios

Q1-50

Q1 Q2

Q2-50

GRAND TOTAL:

100 Marks

SECTION A- APPLICATION BASED QUESTIONS

ALA

Code of Conduct

Shri Harshini Devi - A

[192524104

]

(Name/Reg No) certify that this submission is *my original work and that I have* adhered *to the guidelines specified for this* assessment*. I understand that any violation* of *academic integrity rules* will *result in disciplinary* action.

Signature of the Student:

Ashridharshin

RUBRICS FOR CALCULATION

Criteria

Wt.

Level 4-Exemplary

Understanding of Problem

20%

Clearly understands the problem and accurately identifies all

Good

Context

requirements

Accurately applies

Application of

30%

Concepts

appropriate concepts to *solve*

complex problems

Problem-

Solving

20%

Logical, *well-structured*, and efficient approach

Approach

Accuracy of

Completely correct

20%

Solution

solution with proper justification

Clarity

10%

Clear, well-organized, and properly explained solution

Marks **Evaluation:**

Level 3 Proficient

understanding with minor gaps

Applies relevant concepts correctly with minor **errors**

Mostly logical approach with minor gaps

Mostly correct with minor errors

Understandable with minor clarity issues

Level 2

Developing

Basic

understanding;

some requirements unclear

Applies basic **concepts** with limited accuracy

Basic approach with some inconsistencies

Partially correct solution

Limited clarity and explanation

Poor

Level 1 **Beginning**

understanding or misinterpretation of

the problem

Fails to apply appropriate concepts

Illogical or unclear approach

Incorrect or incomplete solution

Poorly presented

and difficult **to** understand

Understanding of

Question

لمساواة

Problem Context

No.

(20%)

Application of Concepts (30%)

Problem-Solving Approach (20%)

Accuracy of **Solution (20%)**

Clarity *(10%)*

Total Marks

Q1

(50 Marks**)**

Q2

(50 Marks)

Overall **Total (100)**

Faculty In-charge

Course Coordinator

Head of Department

Signature:

Date:

Signature:

Date:

Signature:

Date:

Co1: ASSESSMENT TOOL-1

1) Online Video Streaming Platform:

Recurrence:

T(n) = 2+
(n/2)+

Answer:

n

a=2, b=2, f(n) =

n

logba = nlogs?

2

Case 2 of Master Theorem applies.

Time complexity: O(n log n)

Efficient & scales well for large

user data.

2) E-Commerce Recommendation System:

Compare: $O(n \log n)$ vs $O(n^2)$

Answer:

$O(n \log n)$ grows much slower than $O(n^2)$.

For large datasets, $O(n \log n)$ is more efficient.

It provides better scalability & faster execution.

8) Gaming Engine Optimization:

Compare: $O(n)$, $O(n \log n)$, $O(n^2)$

Answer:

Fastest : $O(n)$

Next: $O(n \log n)$

Slowest : $O(n^2)$

$O(n)$ is the best choice for high-resolution rendering.

34) Online Banking Fraud Detection:

Recurrence:

$T(n) = +(1/2) +)$

Answer:

a=1, b = 2, f(n) =)

Case I of Master Theorem applies.

Time Complexity:

O(logn)

Suitable for real-time fraud detection because it is highly efficient.

61) Smart Parking Management System:

Answer:

Each vehicle is checked once.

Time complexity:

O(n)

Space complexity:

0(1)

The algorithm is efficient for sequential parking allocation.

4) Cloud Storage Processing System:

Recurrence:

T(n) = 3+ (n/3) + n

Answer:

a=3,b=2, f(n) =

nlogs ~ 1.585

n

f(n) is smaller

Case 1 applies.

Time Complexity: 0

(n^1.585)

bad at

Performance grows moderately for large storage systems.

6 Banking Transaction Processing System;

Recurrence:

T(n) = HT(n/o) +n2

Answer:

a=4, b=2, f(n) =n2

n

Log 2 + = n2

Case 2 applies.

Time Complexity: 0

(m2 log n)

Suitable for moderate transaction volumes but becomes slower for very large datasets.

9) Data Backup Optimization System:

Recurrence:

T(n) = ST(n/J) +
√√

Answer:

a=2,b=2, f(n) = √n

log, 2

n

Since √n

grows

slower than

n,

Case 1

applies.

Time Complexity;

O(n)

Efficient for handling large backup datasets.

10) Big Data Analytics Pipeline:

Recurrence:

T(n) = 9 (n−1) +

logn

Answer:

By Substitution,

$T(n) = T(1) + \Sigma logi$

Elogi =

O (nlogn)

Time Complexity: **O(n log n)**

Scales better than quadratic algorithmus.

27) Natural Language Processing Engine:

Recurrence:

T(n) = +(n/a) +

Answer:

Expanding the recurrence gives:

T(n) =n+

n + m2

+n +

み

Sum of the series = 2n.

Time Complexity :

O(n)

Efficient for processing large text datasets.